MASTER'S THESIS · SEMINAR

Optimizing Acoustic Event Detection for Challenging Data and Label Availability

Can semi-supervised learning offset the scarcity of strong labels in sound event detection?

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Deggendorf Institute of Technology · 15.07.2026

Sound reveals the state of a system before vision can

Rattles, squeaks and hisses signal mechanical wear and faults long before a camera sees anything. **Sound Event Detection (SED)** is the task of recognising which acoustic events occur in a signal and at what times.

Condition monitoring

Detect wear, faults and anomalies in machinery from their acoustic signature.

Complements vision

Audio captures events a camera cannot observe, in the dark or out of frame.

Open-ended & hard

Unbounded sources, wide duration range, and overlapping events in real scenes.

The most useful supervision is the scarcest

Not all supervision costs the same. The labels that enable temporal localisation are precisely the ones that are expensive to produce.

STRONG LABELS

Onset + offset per event

Exactly what localisation needs but slow, costly, and prone to annotator disagreement.

SCARCE & EXPENSIVE

WEAK LABELS

Which classes are present

Clip-level presence only no timing. Far cheaper to obtain at scale.

CHEAP

UNLABELLED AUDIO

No annotation at all

Essentially free and abundant the resource this thesis tries to exploit.

ABUNDANT

RESEARCH QUESTION

Can semi-supervised learning exploit weak and unlabelled audio to approach the performance of a fully strongly-supervised SED system on synthetically generated data?

Varying → dataset size · label partitioning · class count · acoustic complexity

From waveform to timed events

SED is a frame-wise, multi-label task: a log-mel spectrogram in, per-frame class activity out, decoded into discrete events with onset and offset times.

01
Raw audio
Mono waveform, 16 kHz

→

02
Log-mel spectrogram
Time × frequency image

→

03
FDY-CRNN
Frame-wise class posteriors

→

04
Timed events
class · onset · offset

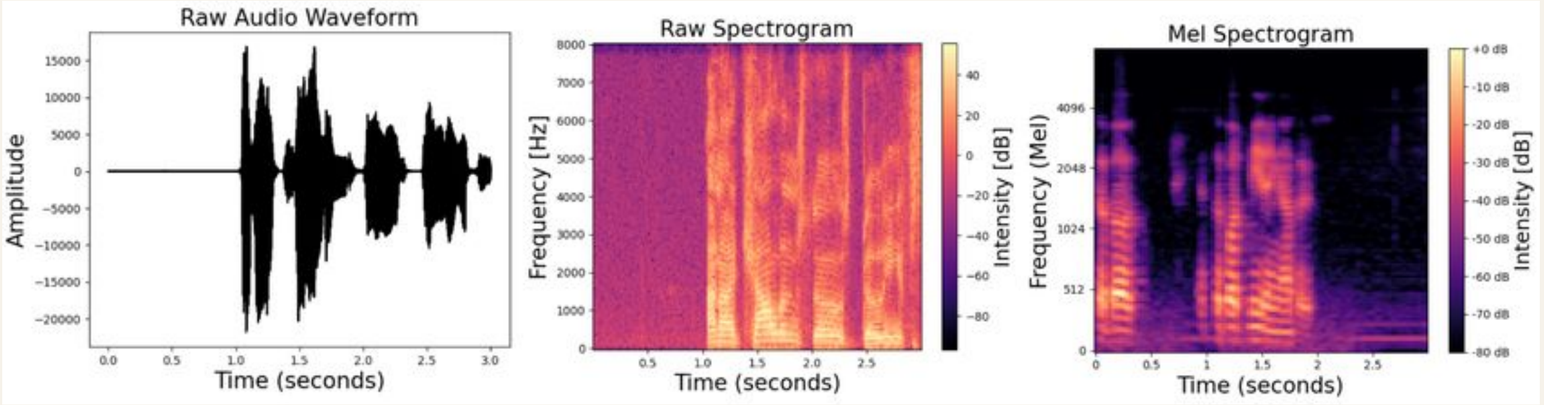


Fig: Graphical representation of conversion from Raw audio signal to Mel-spectrogram

https://www.researchgate.net/figure/Conversion-from-waveform-to-log-Mel-spectrogram-The-waveform-left-shows-amplitude-over_fig1_396787302, 2023.

DATA GENERATION

Three synthetic pipelines, exact timing by construction

Isolated FSD50K events are mixed onto TUT-2017 background noise. Because events are placed programmatically, onset and offset annotations are exact and difficulty is dialled up in three stages.

I.

Single isolated event

One event per clip over background. The easiest regime clean localisation target.

II.

Non-overlapping multi-event

Several events per clip, sequential in time richer per-clip supervision.

III.

Overlapping multi-event

Polyphonic, events coincide in time the realistic, hardest condition.

DATA GENERATION

How each 10-second clip is built

Real isolated events are programmatically mixed onto real background noise under controlled conditions so ground-truth timing is a by-product of construction, not manual work.

EVENTS · FSD50K

Single-label isolated clips

Filtered to one label per clip → approximates strong labels.

Bell Fart Hiss Rattle Squeak

1,081 clips · peak-normalised to 0 dBFS

BACKGROUND · TUT-2017

Real urban / domestic noise

Used only as unlabelled ambience. Random 10 s slice per mixture.

RMS-normalised to −20 dBFS

MIXING ENGINE

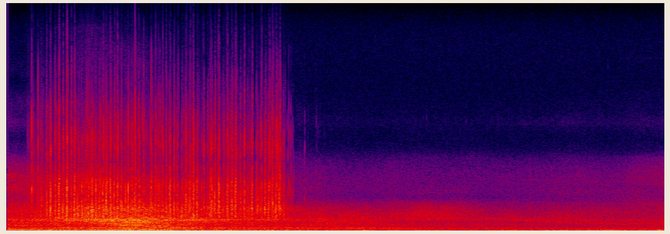
Controlled superposition

1. Random valid placement 100 ms edge buffer, 500 ms gap between events
2. Additive mix event summed onto background sample-by-sample
3. Anti-clip rescale if peak > 1.0, then cast to 16-bit PCM WAV

16 kHz | 10 s clips | ≤ 5 events | seed 42

OUTPUT · STRONG LABELS FOR FREE

Clip + exact annotation



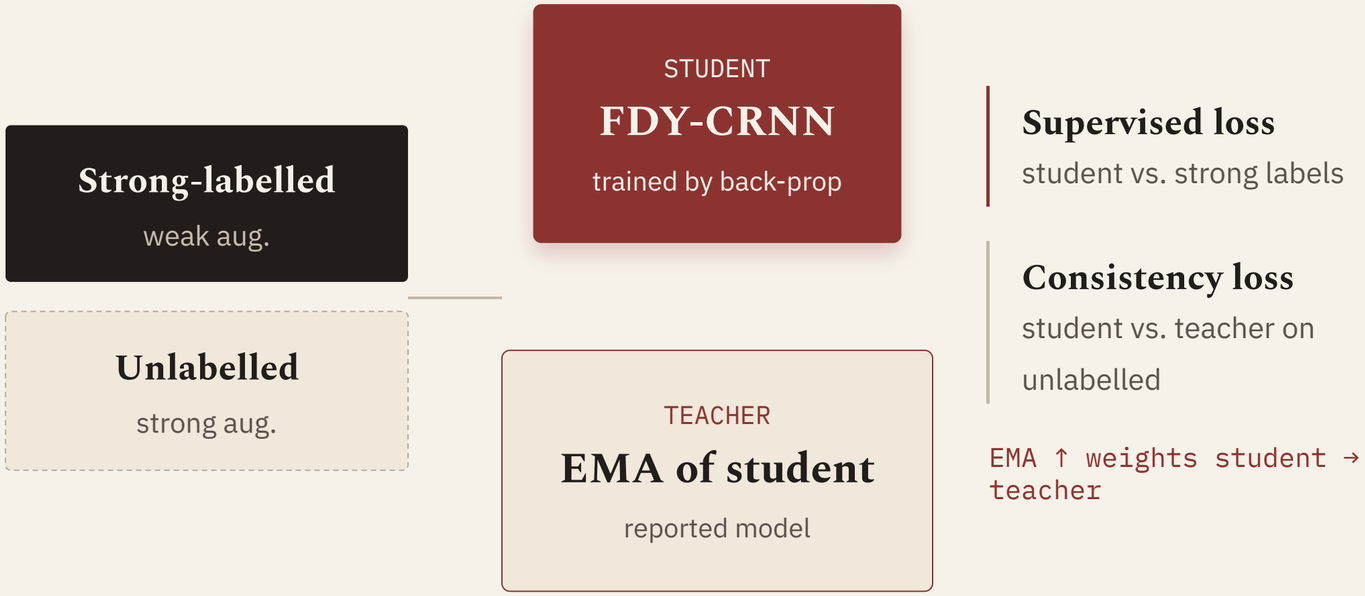
0.0 s 10.0 s

```
mix000004.wav,0.2771,4.9716,Rattle
```

Onset/offset from sample indices → DCASE-compatible CSV. No manual labelling.

A Mean Teacher over a Frequency-Dynamic CRNN

Strong labels drive a supervised loss on the student. Unlabelled clips drive a consistency loss against a teacher whose weights are an EMA of the student a free signal from unlabelled and weak data.



EXPERIMENTAL DESIGN

Fourteen experiments across five groups

Each group holds most factors fixed and sweeps one axis, so effects can be attributed cleanly.

Group	Exp.	Classes	Mixture type	Axis swept
A	E1–E5	3	Single-event	Dataset size & partitioning
B	E6–E7	5	Single-event	Class count (vs. A)
C	E8–E10	5	Non-overlapping	Dataset size
D	E11–E12	5	Overlapping	Source-disjoint eval.
E	E13–E14	5	Overlapping	Higher unlabelled share

Three ways to score a detection

Each metric assumes something different about what counts as “correct” together they show presence, localisation, and threshold-independent behaviour.

SEGMENT-BASED F1

Presence per time-frame



1 s frames each scored independently

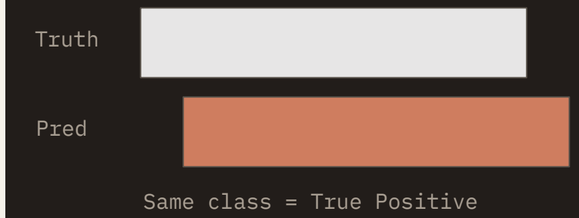
Splits audio into ~1 s frames; each frame is a multi-label decision. Precision/recall/F1 over all frames.

+ robust to boundary jitter - blind to event count & timing

EVENT-BASED F1

PRIMARY

Discrete events, matched



Each event = onset, offset, class. Matched to truth by temporal overlap (or onset collar).

+ true localisation of instances - strictest; offsets hard

PSDS

Threshold-free, area-based

Integrates over all operating points no single threshold to tune or bias the score.

PSDS1 - strict localisation PSDS2 - tagging-oriented, tolerant

REFERENCES & EVALUATION

What we compare against and how we score it

THREE REFERENCE SYSTEMS

Supervised strong-label baseline

Upper reference on the same overlapping data.

ATST-SED (fine-tuned transformer)

Transfer from large-scale external pre-training.

Event-to-Background Ratio sweep

One fixed model across quieter and quieter events.

METRICS

Event-based macro F1

Primary. Matches whole events by onset/offset; each class weighted equally.

PSDS1 precise temporal localisation **Segment F1** frame-level presence macro + micro averaging reported

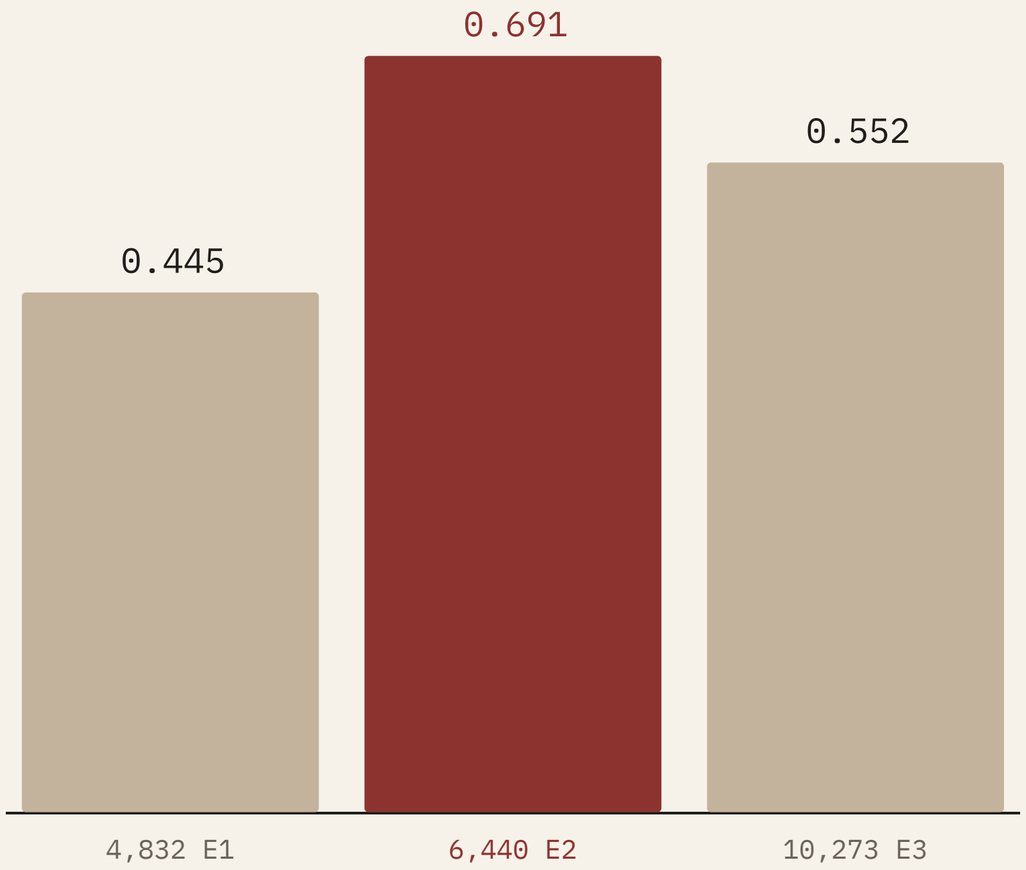
Results

Teacher metrics, reported as mean ± standard deviation over repeated runs.
Event-based macro F1 is primary.

GROUP A · THREE-CLASS SINGLE-EVENT

More data isn't always better and unlabelled trades poorly

TEACHER EVENT-F1 VS. DATASET SIZE



Peak at 6,440 mixtures, then decline: at fixed partitioning a larger unlabelled pool did not help under fixed consistency weight and EMA decay.

PARTITIONING AT FIXED 8,050 MIXTURES

Exp.	Unlab.	Event-F1	PSDS1
E5	35%	0.481	0.662
E4	40%	0.418	0.502

Raising the unlabelled share from 35% to 40% lowered every metric.

Strong labels the model was using were removed; consistency did not compensate.

GROUP B · FIVE-CLASS SINGLE-EVENT

More classes sharply raise difficulty

MATCHED PAIR · SAME PARTITIONING

0.445 → **0.237**

E1 · 3 classes

E6 · 5 classes

−47% teacher event-F1

Each added class dilutes a fixed strong-label budget across more decision boundaries.

The macro average also penalises weak per-class recall harder, so rare or confusable classes drag the score down more visibly than in the three-class setting.

GROUP C · FIVE-CLASS NON-OVERLAPPING

With richer clips, scaling data helps again

Multi-event clips give more supervision per clip so more data now helps monotonically.

Two mechanisms compound: each clip contains several labelled events, and the sequential (non-overlapping) layout keeps them individually resolvable.

DIRECTION OF EFFECT

- ↗ Larger dataset → higher event-F1
- ↗ Single- → multi-event raises scores
- ≠ Contrast: Group A was non-monotonic

GROUPS D & E · OVERLAPPING MULTI-EVENT

Honest evaluation lowers the ceiling and more unlabelled still hurts

GROUP D

A source-disjoint test split exposes leakage

Ensuring no sound source appears in both train and test lowered absolute event-based scores showing earlier numbers were partly inflated by source-level train test overlap. The result: a lower but more credible performance ceiling.

GROUP E

Raising unlabelled 45% → 55% degrades accuracy

At both 12,000 and 18,000 mixtures, reallocating strong labels to the unlabelled pool reduced every reported metric the same trade-off seen in Group A, now confirmed in the hardest regime.

E12 teacher · event-F1 macro = 0.358

REFERENCE · SUPERVISED BASELINE

Strong labels alone already cover the classes

0.571

event-F1 · 40% strong labels only

vs. 0.358 semi-supervised (E12)

On this data the binding constraint is strong-label coverage not the amount of unlabelled audio.

The 40% strong partition is large enough that every source event recurs many times, so it already provides sufficient class coverage. The unlabelled pool holds little the strong labels don't so trading one for the other only removes supervision.

REFERENCE · TRANSFER LEARNING

External pre-training beats in-domain unlabelled data

Fine-tuning ATST-SED on the identical E12 configuration improves every metric over the Mean Teacher system with the largest gains on the localisation-sensitive scores.

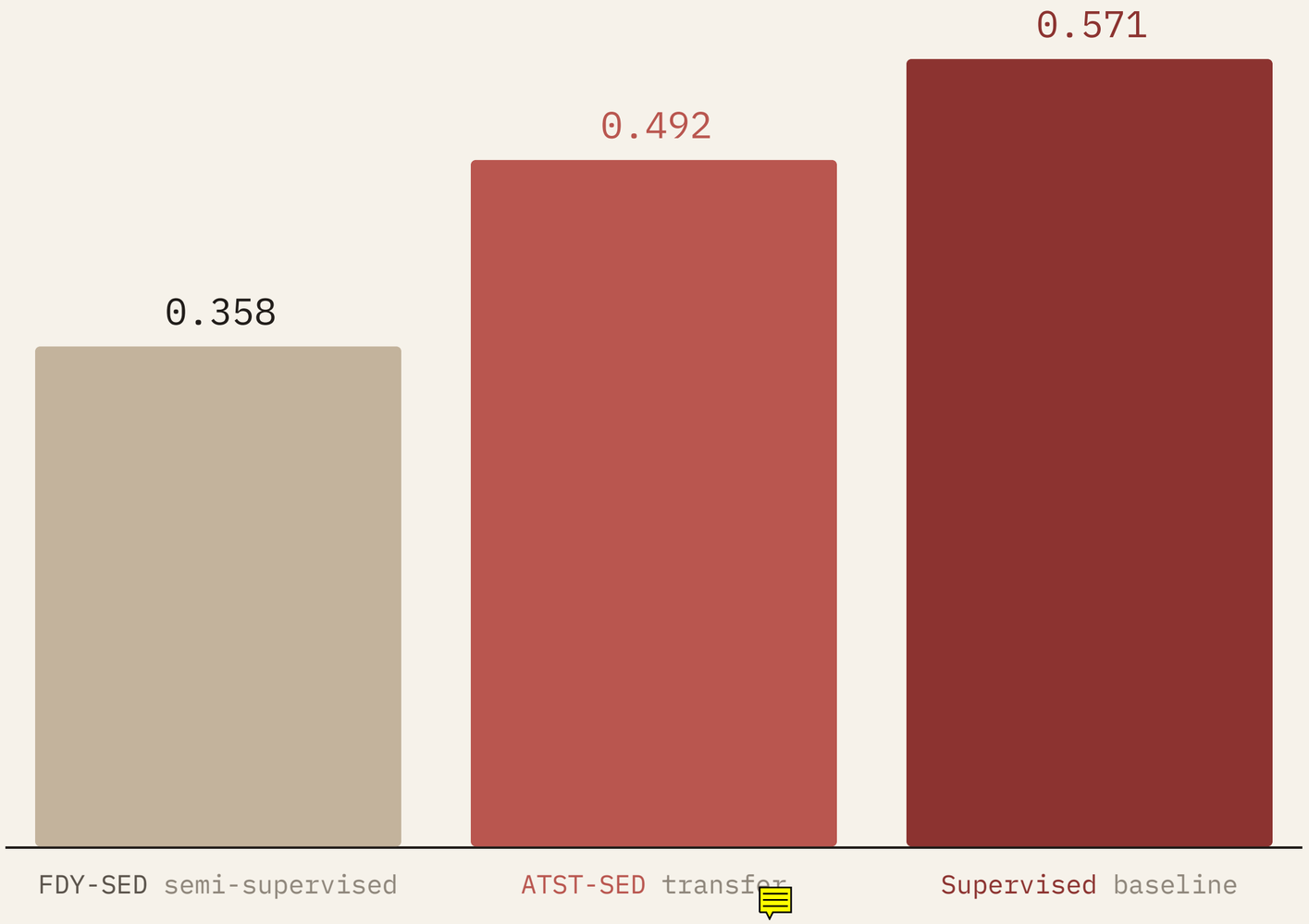
E12 configuration	Event-F1 (macro)	PSDS1	Segment-F1
FDY-SED semi-supervised	0.358	0.227	0.793
ATST-SED fine-tuned	0.492	0.367	0.810
△ improvement	+0.134	+0.140	+0.017

Big gains on event-F1 and PSDS1 both reward precise timing show transferred frame-level representations supply localisation the in-domain Mean Teacher cannot recover. Segment-F1 was already near-saturated for both.

CROSS-MODEL COMPARISON · IDENTICAL E12 DATA

Transfer narrows most of the gap to full supervision

EVENT-BASED MACRO F1



≈86%

of supervised event-F1 recovered by ATST-SED...

40%

...while still using only 40% of the pool as strong labels.

ROBUSTNESS · EVENT-TO-BACKGROUND RATIO SWEEP

As events grow quieter, timing is lost before presence



Precise onset/offset estimation is the **first capability lost** as events recede; frame-level presence survives far longer (0.487 even at -15 dB).

CONCLUSION

The result didn't meet the expectations.

NULL · IN-DOMAIN SSL

Trading strong labels for unlabelled data can help but in limits

Across every group, raising the unlabelled share degraded performance, and the Mean Teacher system did not match the supervised baseline. The binding constraint was strong-label coverage not unlabelled availability.

0.358 SSL · 0.571 supervised

POSITIVE · TRANSFER

External pre-training is the more promising route

ATST-SED pretrained model gave a increase in scores compared to semi-supervised model.

0.358 → 0.492 event-F1

FUTURE WORK

Where this points next

01

Stronger use of unlabelled data

Confidence-thresholded pseudo-labelling, curriculum and self-training beyond weak Mean Teacher consistency.

02

Combine pre-training with SSL

Initialise from a pre-trained backbone, then run the semi-supervised pipeline on top the two are complementary.



THANK YOU

Questions & discussion

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Deggendorf Institute of Technology · 15.07.2026